

Customer Shopping Behavior Analysis

Turning 3,900 transactions into a clear read on who buys, what drives spend, and where to focus next

Python · PostgreSQL · Power BI · Leadership readout

DATASET 1.0

3,900 transactions, 18 fields, one goal: actionable segments.

Demographics

Age, gender, location, and subscription status identify who is buying.

Purchase details

Item, category, amount, season, size, and color describe what they bought.

Behavior

Discounts, promo use, purchase frequency, review ratings, and shipping type show how they buy.

3,900

Transactions analyzed

18 fields

3 field groups

1 customer view

Clean data first: from raw rows to a working model.

01 Loaded and profiled the dataset in Python

pandas gave the first read on distributions, gaps, and field types.

02 Imputed 37 missing review ratings

Each gap took its own category's median rating.

03 Standardized columns and engineered two features

Columns moved to snake_case; age_group and purchase_frequency_days were added.

04 Dropped the redundant promo field

promo_code_used matched discount_applied on a consistency check.

05 Loaded the clean table into PostgreSQL

One structured model became the base for every query that follows.

Ten business questions, answered directly in SQL.

With clean data in PostgreSQL, we queried straight for decision-ready answers.

No.	Question	No.	Question
01	Revenue by gender Where spend concentrates	04	Top-rated products Consistent performers
02	Revenue by age group Which cohorts pay	05	Shipping spend Standard vs. express
03	High spenders using discounts Margin left on the table	06	Subscriber value Members vs. non-members

Six of the ten questions carry the decisions in this readout; the remainder cover category, season, and size cuts.

FINDING 3.1

Subscribers and top segments carry outsized value.

Subscription status and purchase history are the clearest lines between casual and high-value customers. Subscribers show higher average spend and higher total revenue than non-subscribers.

Segment rule

Customers with more than five purchases are more likely to hold an active subscription.

CUSTOMER TIERS BY PURCHASE HISTORY

Loyal

Highest spend and the strongest subscription rate.

Returning

Repeat buyers with room to move up a tier.

New

First purchases; subscription is the next step.

Discounts move volume, but not always the right customers.

SPEND

High spenders still use discounts

A meaningful share of above-average customers redeem a discount. Margin is on the table where demand already exists.

PRODUCTS

A small set is discount-dependent

These items rarely sell at full price. Weak full-price demand, not a pricing accident.

SHIPPING

Express buyers spend differently

Average purchase amount differs between standard and express, pointing to a distinct express segment.

FINDING 4.1

A clear set of products leads on rating and repeat purchase.

The top five products by average review rating hold up as consistent performers. Within each category, the top three define the assortment worth protecting.

Implication

These products are the natural anchors for merchandising and campaign focus.

Selection logic

05

Top products by average rating

03

Leaders per category

02

Signals used: rating and repeat purchase

02

DASHBOARD 5.0

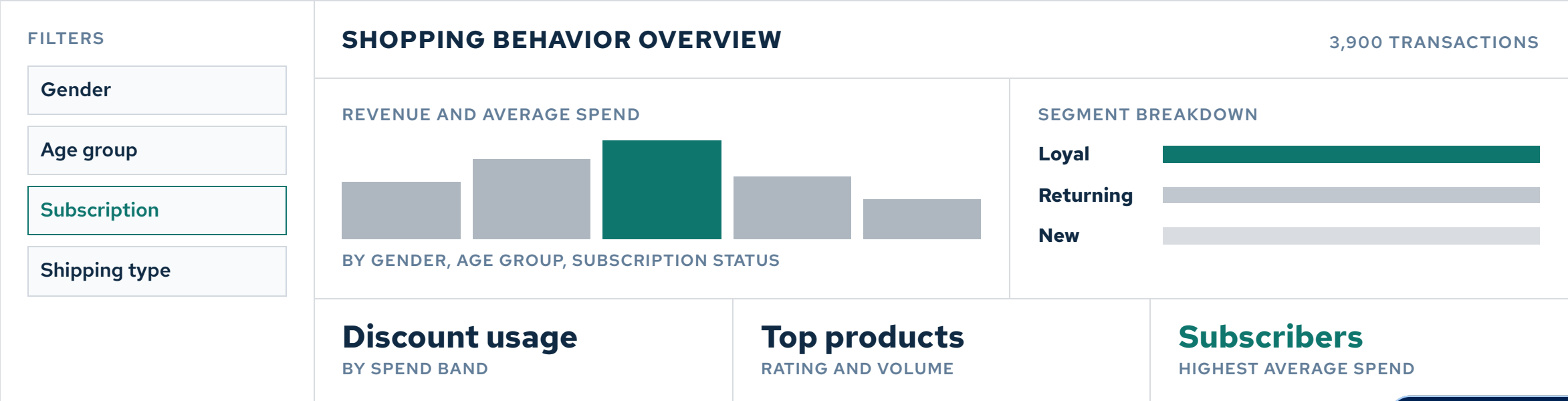
From findings to a live view: the Power BI dashboard

Spend · segments · [products](#)

One dashboard brings spend, segments, and products into one view.

EXHIBIT 5.1

The SQL findings were built into an interactive Power BI dashboard so stakeholders can explore shopping behavior without writing a query.



Five moves to grow revenue and protect margin.

RECOMMENDATION

WHY IT MATTERS

Boost subscriptions

Subscribers already spend and return more.

Build loyalty programs

Moves repeat buyers into the high-value Loyal segment.

Review discount policy

Protects margin from high spenders who don't need the discount.

Sharpen product positioning

Puts top-rated, best-selling items at the front of campaigns.

Target by segment

Focuses marketing spend on high-revenue age groups and express-shipping users.